

Recalling version: Functional Analysis Final Examination B

2024-2025-2

120 minutes

Name: _____
Student ID: _____
Signature: _____

- (20 points) Let X be a Banach space and $\{x_n\} \subset X$.
 - Show that if $\{x_n\}$ is weakly convergent, then the convergence is unique.
 - Show that if $x_n \rightharpoonup x$ and $f_n \rightarrow f$ for $\{f_n\} \subset X^*$ and $f \in X^*$, then $f_n(x_n) \rightarrow f(x)$.
 - Show that if X is Hilbert, $x_n \rightharpoonup x$ and $\|x_n\| \rightarrow \|x\|$, then $x_n \rightarrow x$.
- (20 points) Let $C([-1, 1])$ be the space of real continuous functions with sup norm. $T : C([-1, 1]) \rightarrow \mathbb{R}$, $Tf = \int_0^1 f - \int_{-1}^0 f$.
 - Show that $T \in \mathcal{B}(C([-1, 1]))$.
 - Compute the operator norm $\|T\|$.
 - Is $\|T\|$ attaining? That is, is there exists $\phi \in C([-1, 1])$ such that $|T\phi| = \|T\|$?
- (20 points) Let $\mathcal{H}$ be Hilbert.
 - $F \subset \mathcal{H}$ is closed subspace, show that $\mathcal{H} = F \oplus F^\perp$.
 - Give an example of Hilbert space $\mathcal{H}$ and subspace K , such that $\mathcal{H} \neq K \oplus K^\perp$.
 - If $\mathcal{H}$ is separable, show that $\mathcal{H} \cong l^2$ with an isometric isomorphism.
- (20 points) Let $\mathcal{H}$ be Hilbert, $T \in \mathcal{B}(\mathcal{H})$ and T is compact.
 - State the definition of $\sigma(T)$.
 - Show that if $\mathcal{H}$ is infinite-dimensional, then $0 \in \sigma(T)$.
 - Show that $\sigma(T)$ has at most one accumulation point.
- (20 points) Let $X \subset C([- \pi, \pi])$ a closed subspace containing all f such that $f(-\pi) = f(\pi)$, $\{e_n(t)\} \subset X$ for $e_n(t) = \frac{e^{int}}{\sqrt{2\pi}}$, $\{\alpha_n\} \subset \mathbb{C}$, Let $\hat{f}(n) = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \frac{f(t)e^{-int}}{\sqrt{2\pi}} dt$. Suppose $\forall f \in X$, there exists unique $g \in X$ such that $\forall n \in \mathbb{Z}$, $\hat{g}(n) = \alpha_n \hat{f}(n)$. Define $Tf = g$.
 - Show that $T \in \mathcal{B}(X)$.
 - Show that $Te_n = \alpha_n e_n$, and $\{\alpha_n\}$ is bounded.
 - Show that there exists $\phi \in X^*$, such that $\phi(e_n) = \alpha_n$.

*End of Recalling version.
Good luck!*